Energy M&A Scraper Technical Deep-Dive

Libraries, Methods, AI Architecture & Cost Analysis

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1. Libraries, Scraping Methods & Workflow

Core Libraries Used

Parameter	Value / Description
aiohttp / cloudscraper	Async HTTP requests for high-concurrency fetching. Bypass basic bot detection.
Playwright (	

Workflow & Libraries Flowchart

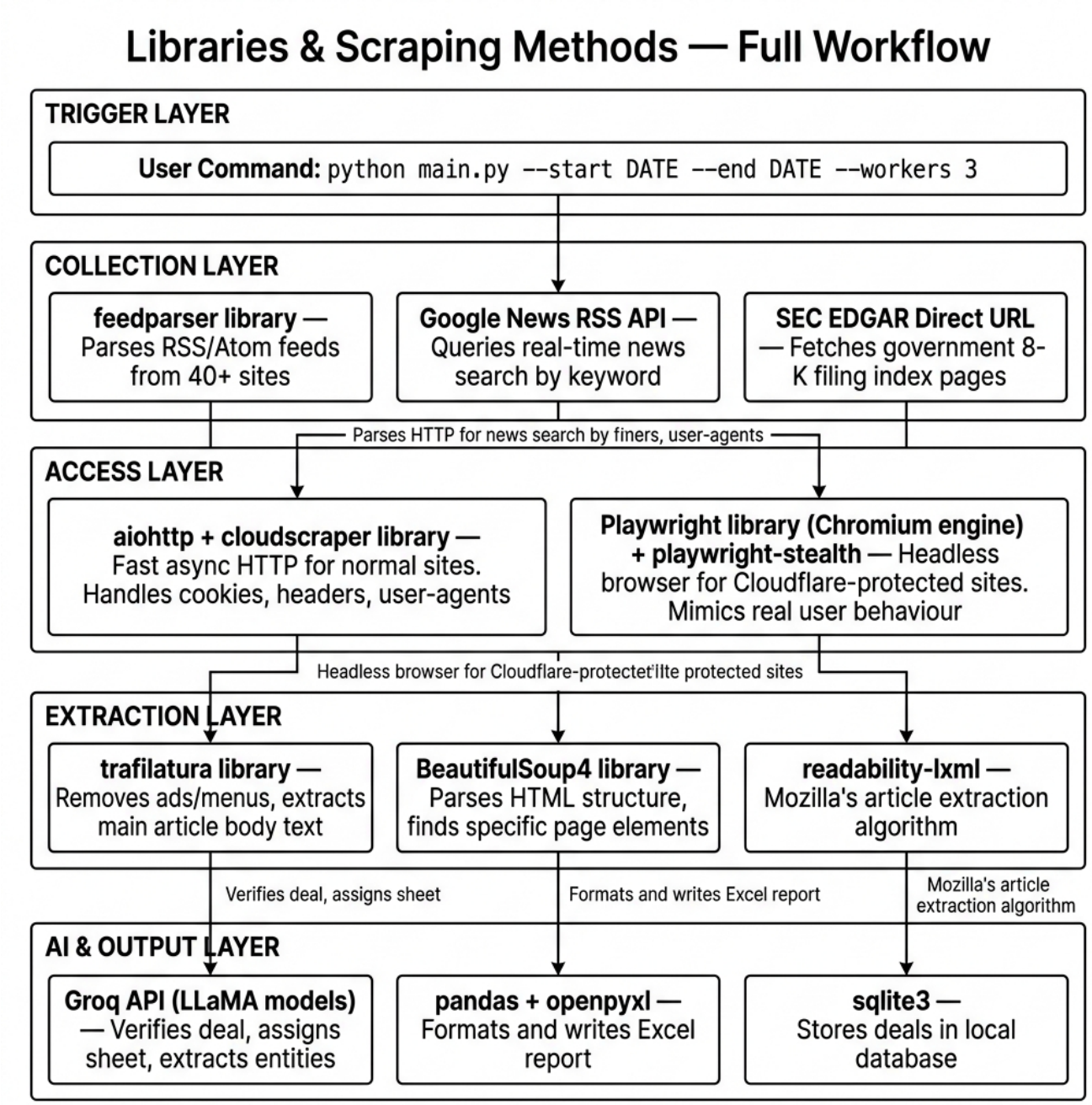

Figure: Scraping libraries used across the multi-layer pipeline

The workflow consists of layer-by-layer processing:

-> Trigger: python main.py inputs the search window.

-> Collection: RSS, Google News, and SEC endpoints are queried.

-> Access: Stealth browsers handle security challenges automatically.

-> Extraction: Content is stripped of ads and navigation menus.

-> Intelligence: AI validates, classifies, and saves results to DB and Excel.

2. AI API Role, Prompts & Scraping Interaction

Role of AI in the Pipeline

The AI (Groq API) acts as a high-speed analyst filtering through thousands of potential articles.

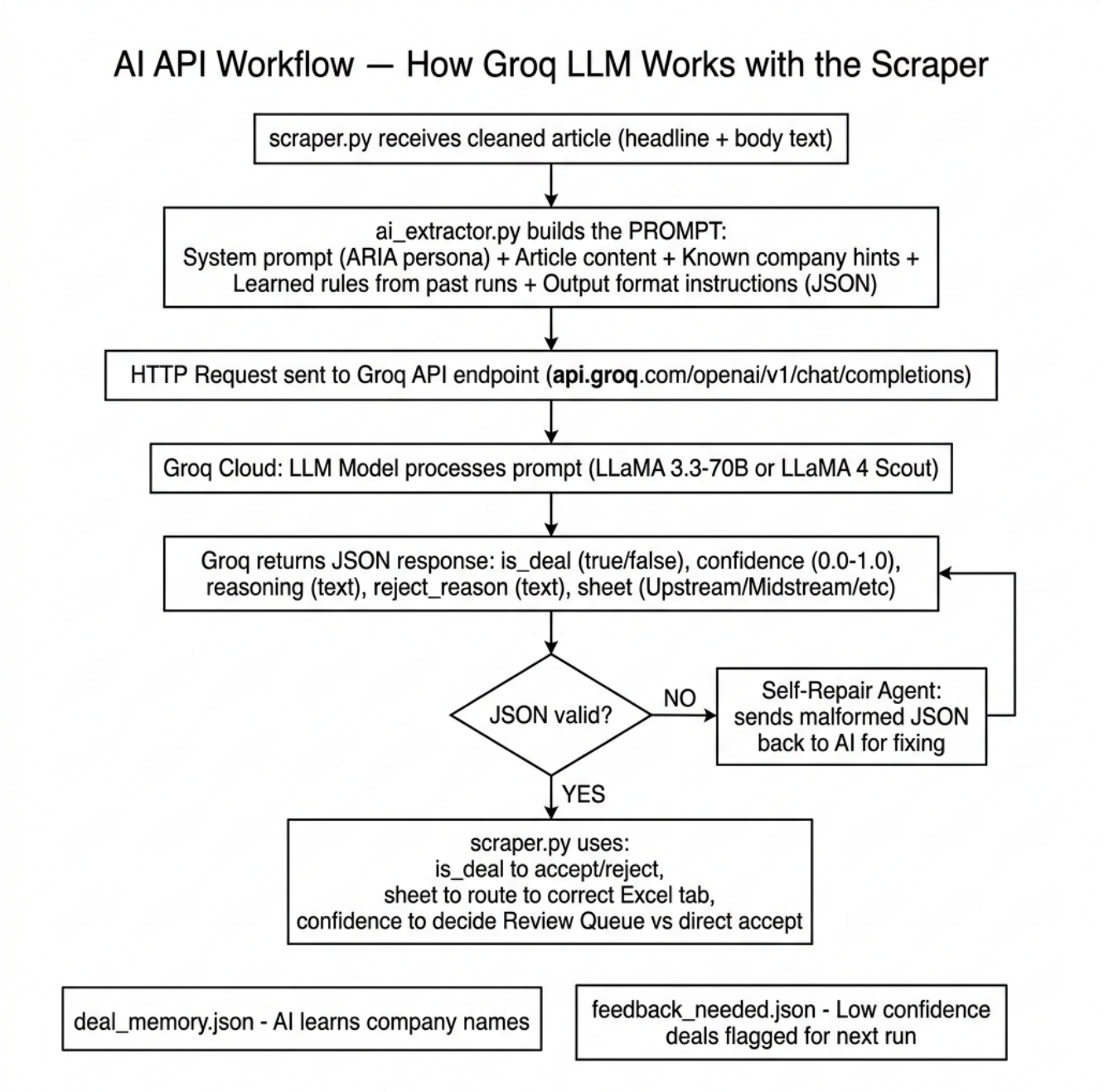

Figure: How the AI interacts with the core scraper engine

Prompts Provided to AI

System Prompt (Analyst Identity):

> "You are ARIA, a senior energy analyst. Your role is M&A verification. You extract structured data only if a transaction is confirmed."

User Prompt (Data Task):

"Analyze the following article. Is this an M&A deal? If yes, provide buyer, seller, asset, and value in JSON format."

-> Structured Extraction: Extracting Buyer/Seller names specifically.

-> Classification: Assigning to sheets (Upstream, Midstream, etc.).

-> Normalization: Converting mixed currencies to normalized USD values.

3. AI Feedback & Self-Learning System

How Feedback Works

-> Confidence Scoring: AI assigns a 0.0-1.0 confidence score to every deal.

-> Automation: Deals with score < 0.8 are flagged in feedback_needed.json.

-> Self-Correction: On the next run, the system prioritizes re-reviewing low-confidence items.

-> Memory: deal_memory.json stores confirmed acquirers to boost future accuracy.

4. Monthly Cost Analysis (Budgeting)

Parameter	Value / Description
Groq AI API (Current)	$0.00 / Month (Fits within free tier limits)
Google Vert	

Assuming 1 run per day on 84 sites for 30 days:

-> Total articles scanned: ~15,000 / month

-> AI verification calls: ~6,000 / month

-> Total Monthly Budget: $0 to $5 (Extremely efficient)

5. Next Steps for Scraping & AI Efficiency

-> Implement Proxy Rotation to prevent IP blocks from sensitive sites.

-> Move from purely static config to a dynamic sitemap parser.

-> Switch from Llama-3-70B to Llama-3.1-405B for ultra-complex deal extraction.

-> Integrate a semantic deduplication layer using vector embeddings.

6. AI Intelligence Roadmap: How Smart Can It Be?

-> Predictive Alerts: AI predicts future deals based on partnership activity.

-> Valuation Multiples: Auto-calculating deal multiples vs. public peers.

-> Auto-Research: AI web-searches for missing values (e.g., deal price not in report).

7. Deployment Options for Office Laptops

Parameter	Value / Description	Streamlit We
CLI (Current)	Fastest execution. Requires PowerShell knowledge.	

8. Google Vertex AI: Enterprise Upgrade

Switching to 'Gemini 1.5 Flash' on Vertex AI provides:

-> Enterprise Security: Full SOC2/HIPAA compliance.

-> Data Privacy: Your data is NOT used to train Google models.

-> Ultra-low latency: Sub-second response times for faster runs.

-> Credit Utilization: Flexible pay-as-you-go credit billing.